

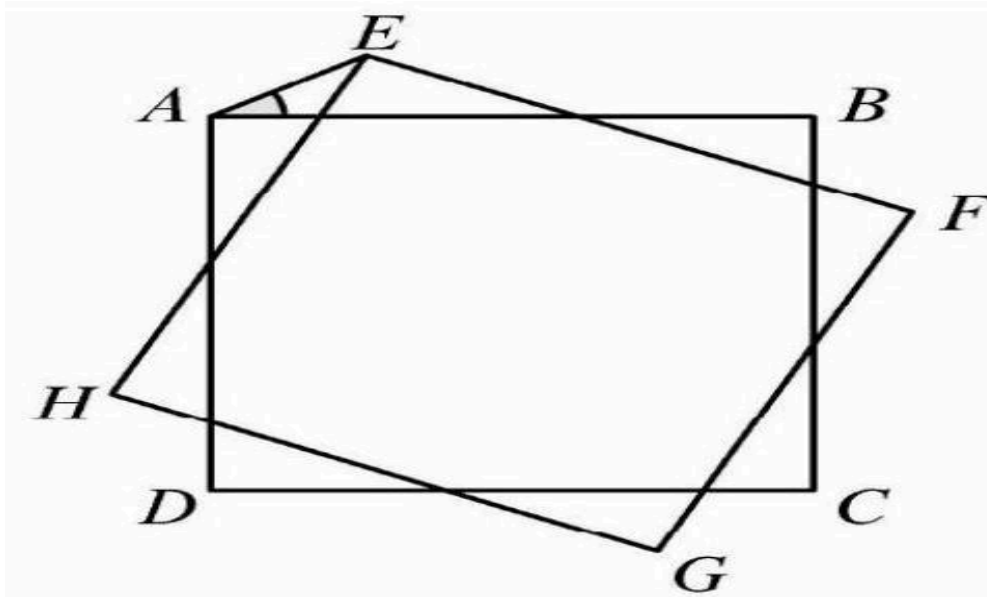
2023 AMC 10B Problem 7 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 7

Problem:

Square $ABCD$ is rotated 20° clockwise about its center to obtain square $EFGH$, as shown below. What is the degree measure of $\angle EAB$?



Answer Choices:

- A. 24°
- B. 35°
- C. 30°

D. 32° E. 20° **Solution:**

Let O be the common center of both squares. Because $AO = EO$, triangle $\triangle AOE$ is isosceles with $\angle AOE = 20^\circ$. This means $\angle EAO = \frac{1}{2}(180^\circ - 20^\circ) = 80^\circ$. Combined with $\angle BAO = 45^\circ$, this yields

$$\angle EAB = \angle EAO - \angle BAO = 80^\circ - 45^\circ = \text{(B)} \boxed{35^\circ}$$

OR

The vertices $ABCDEFGH$ divide the circumscribing circle into 8 arcs. Four are size 20° , so the other four are $\frac{360^\circ - 4 \cdot 20^\circ}{4} = 70^\circ$. Angle $\angle EAB$ is half the measure of the central angle $\angle EOB$, where O is the center of the circumscribing circle, so $\angle EAB = \frac{70^\circ}{2} = \text{(B)} \boxed{35^\circ}$.

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